Review: Double Integrals

MATH 321 – Mathematical Statistics I

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Iterated Integrals

An **iterated integral** is evaluated from the inside out:

$$\int_a^b \int_c^d f(x,y)\,dy\,dx = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx.$$

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The differentials are read from the inside out: the innermost differential (dy) goes with the innermost integral sign ($\int_c^d$).

Fubini's Theorem

Theorem (Fubini): If f is continuous on the rectangle $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy.$$

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Over a rectangle, reversing the order of integration is easy: **the bounds just travel with their variable**.

Example 1: A Rectangular Region

Evaluate $\iint_R 6xy^2\,dA$, where $R = \{(x, y) : 0 \leq x \leq 2,\ 0 \leq y \leq 1\}$.

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Integrating y first:

$$\begin{aligned}\int_0^2 \int_0^1 6xy^2\,dy\,dx &= \int_0^2 \Big[2xy^3\Big]_{y=0}^{y=1} dx \\ &= \int_0^2 2x\,dx \\ &= \Big[x^2\Big]_0^2 = 4.\end{aligned}$$

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Integrating x first:

$$\begin{aligned}\int_0^1 \int_0^2 6xy^2\,dx\,dy &= \int_0^1 \Big[3x^2y^2\Big]_{x=0}^{x=2} dy \\ &= \int_0^1 12y^2\,dy \\ &= \Big[4y^3\Big]_0^1 = 4.\checkmark\end{aligned}$$

A Shortcut for Products

If the region is a rectangle **and** the integrand factors as $f(x, y) = g(x)\, h(y)$, then

$$\int_a^b \int_c^d g(x)\, h(y)\, dy\, dx = \left(\int_a^b g(x)\, dx \right) \left(\int_c^d h(y)\, dy \right).$$

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Warning: This only works when *both* conditions hold. If the bounds depend on x or y, the integral does **not** split.

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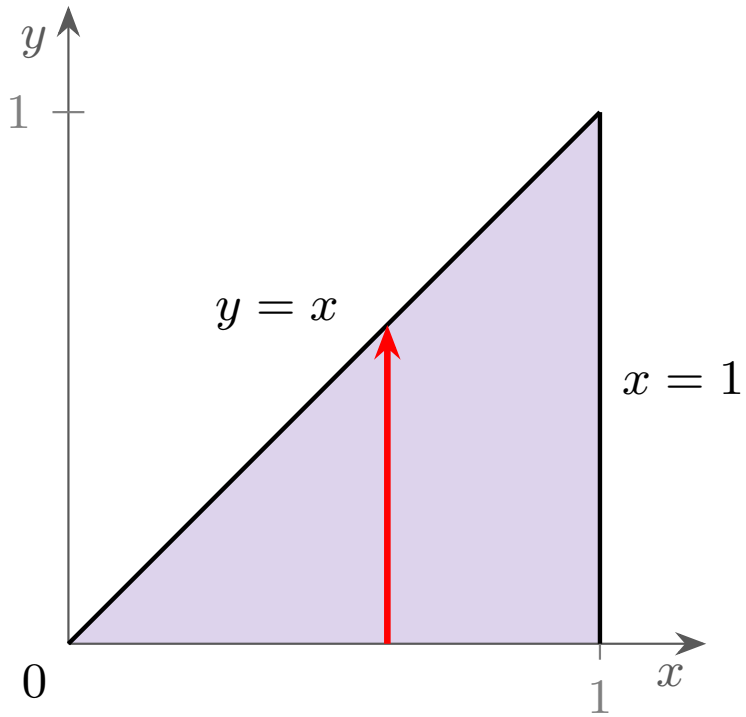
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Think in slices: with $dy\,dx$, fix x and draw a **vertical** line through the region. Its bottom and top give the y-bounds. With $dx\,dy$, fix y and draw a **horizontal** line. Its left and right ends give the x-bounds.

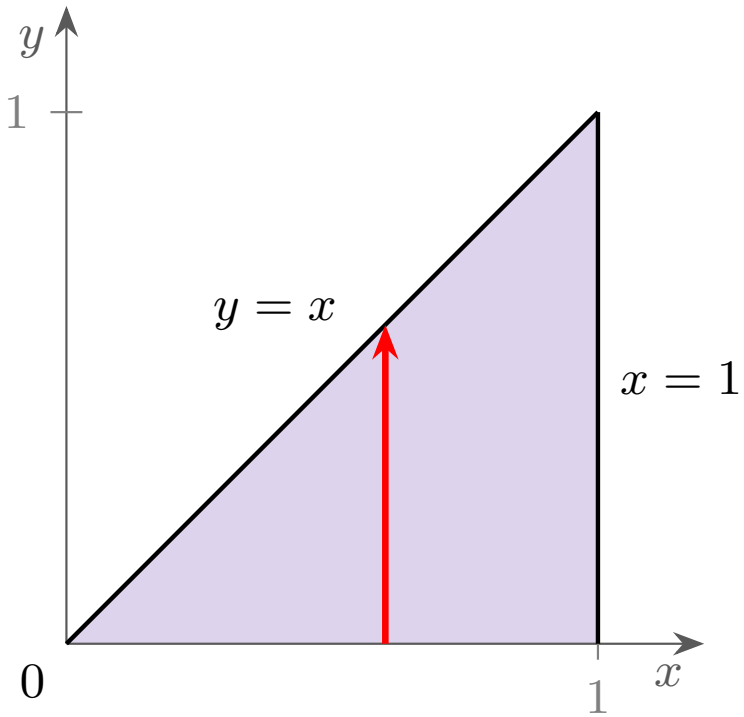
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Vertical slice at x: y runs from 0 up to x.

$$\int_0^1 \int_0^x 6y\,dy\,dx = \int_0^1 \Big[3y^2\Big]_{y=0}^{y=x} dx$$

$$= \int_0^1 3x^2\,dx$$

$$= \Big[x^3\Big]_0^1 = 1.$$

Since $f \ge 0$ and it integrates to 1, f is a valid joint density.

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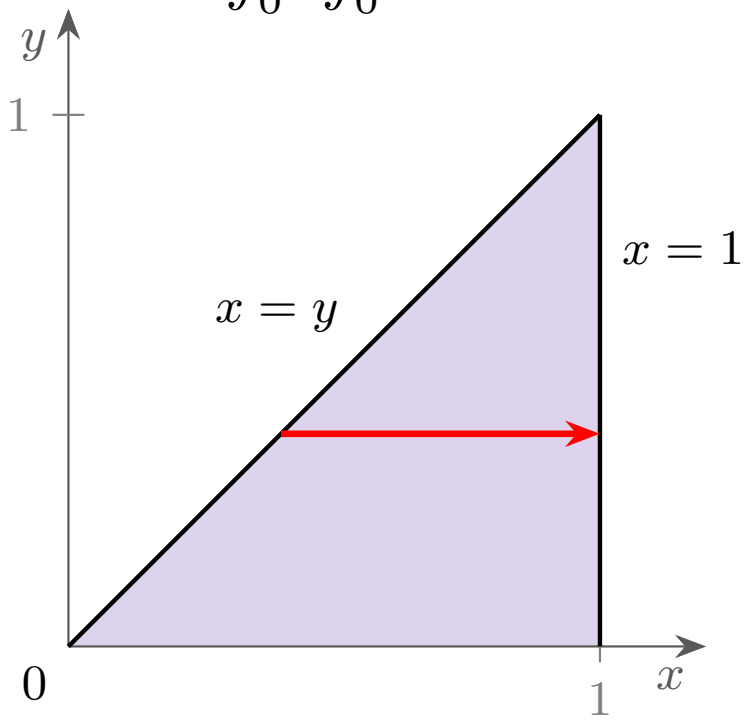
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4. **Find the new inner bounds:** fix the outer variable, draw a slice across the region, and see which curves it enters and exits through. Solve those curves for the inner variable.
5. **Check for a split:** if the entry or exit curve changes partway through, break the integral into pieces.

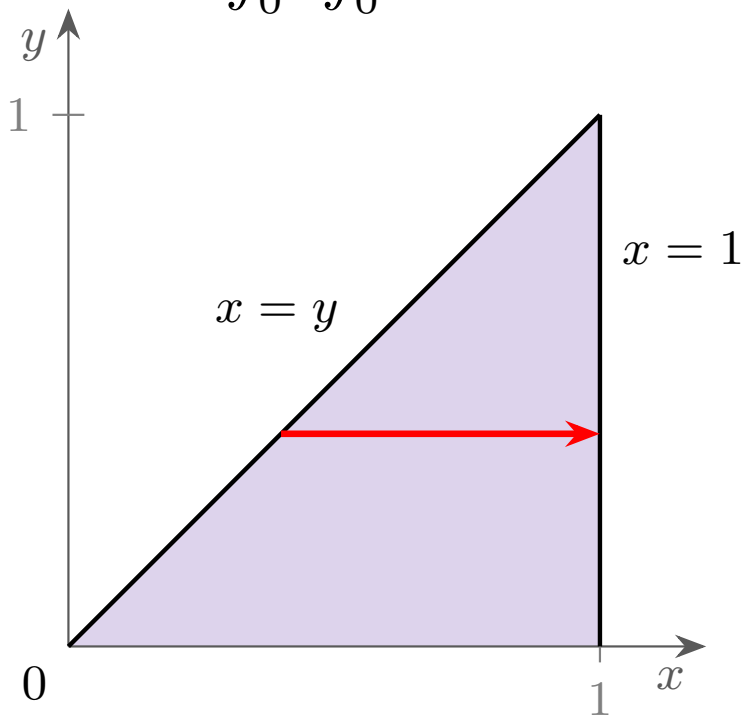
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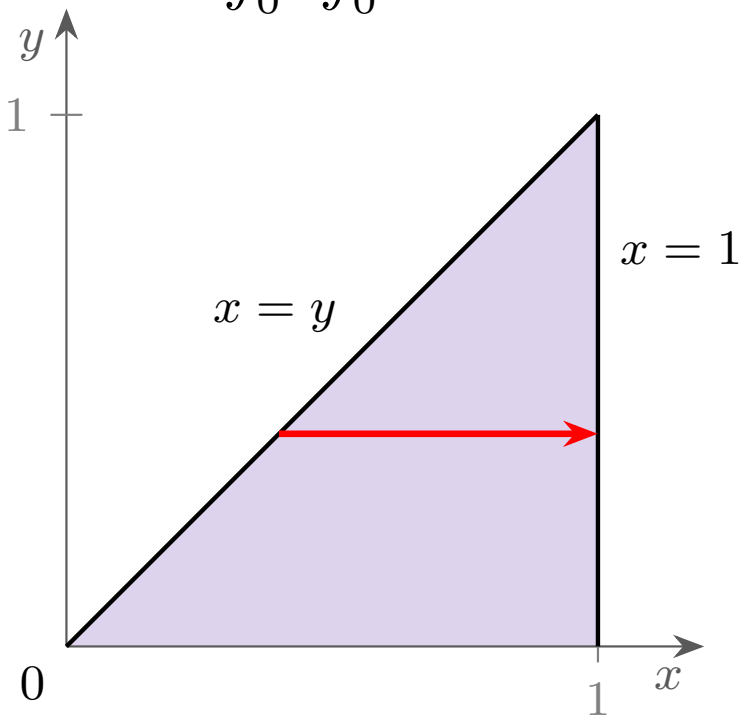
Rewrite $\int_0^1 \int_0^x 6y \, dy \, dx$ in the order $dx \, dy$.



- y ranges over $[0, 1]$ in the region.
- Horizontal slice at y: enter at $x = y$, exit at $x = 1$.

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$$\int_0^1 \int_y^1 6y \, dx \, dy = \int_0^1 \Big[6xy\Big]_{x=y}^{x=1} dy$$

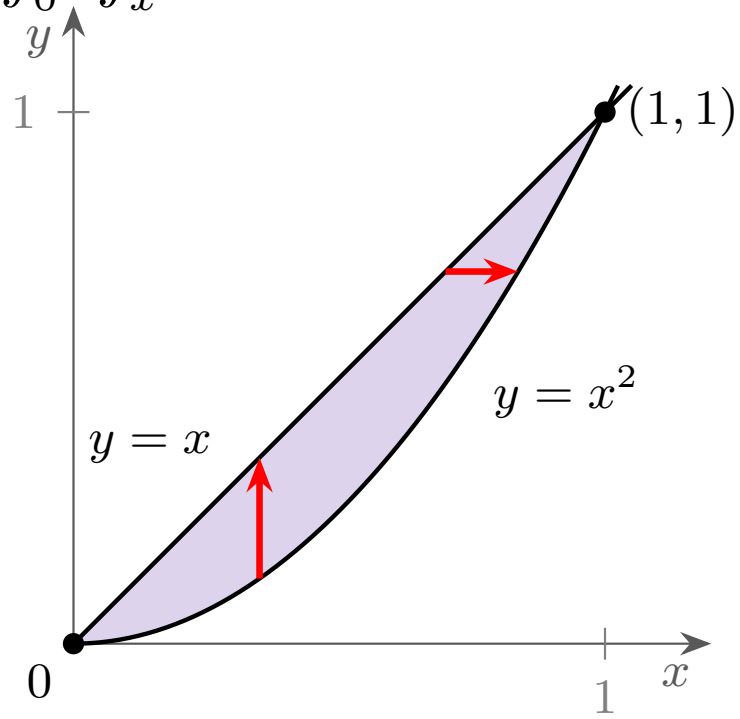
$$= \int_0^1 (6y - 6y^2) \, dy$$

$$= \Big[3y^2 - 2y^3\Big]_0^1 = 1. \checkmark$$

Example 3: Between Two Curves

Reverse the order of integration and evaluate both ways:

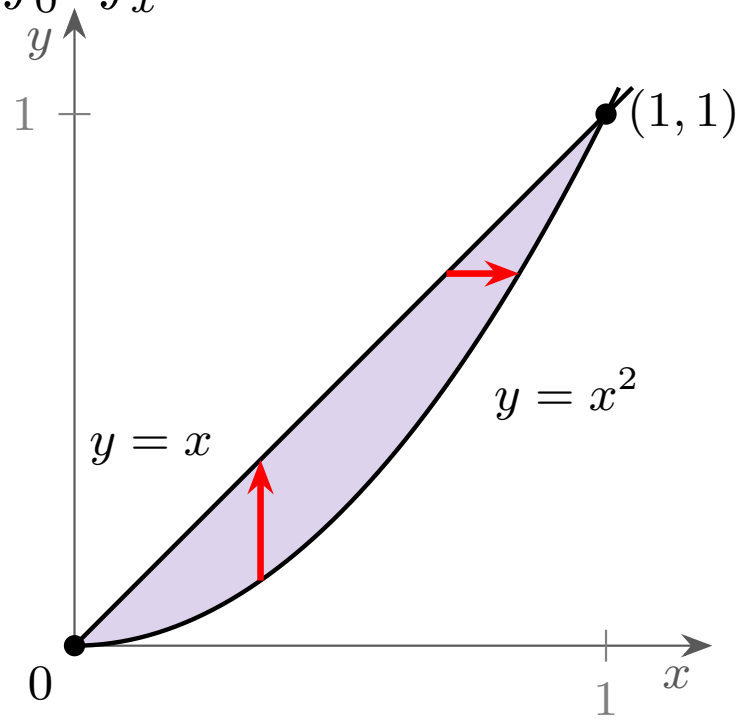
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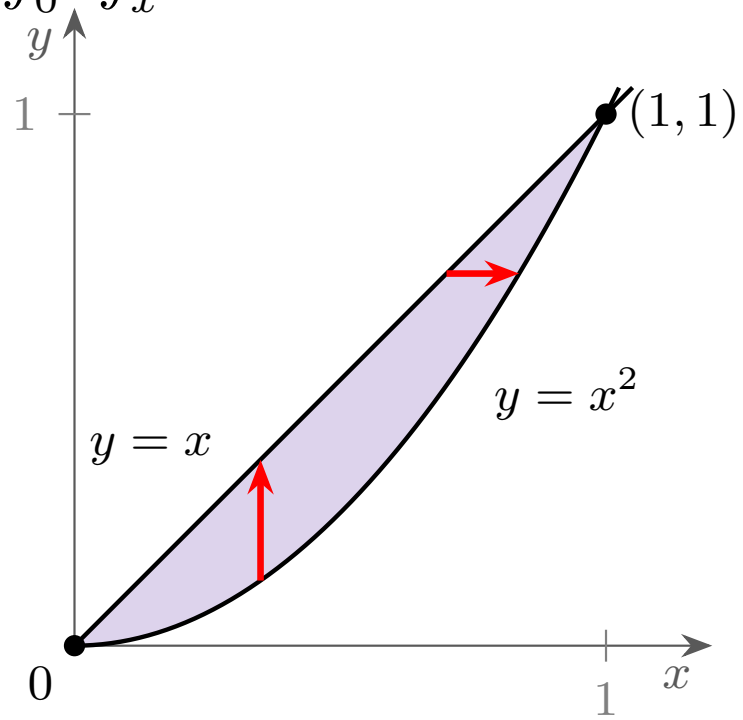


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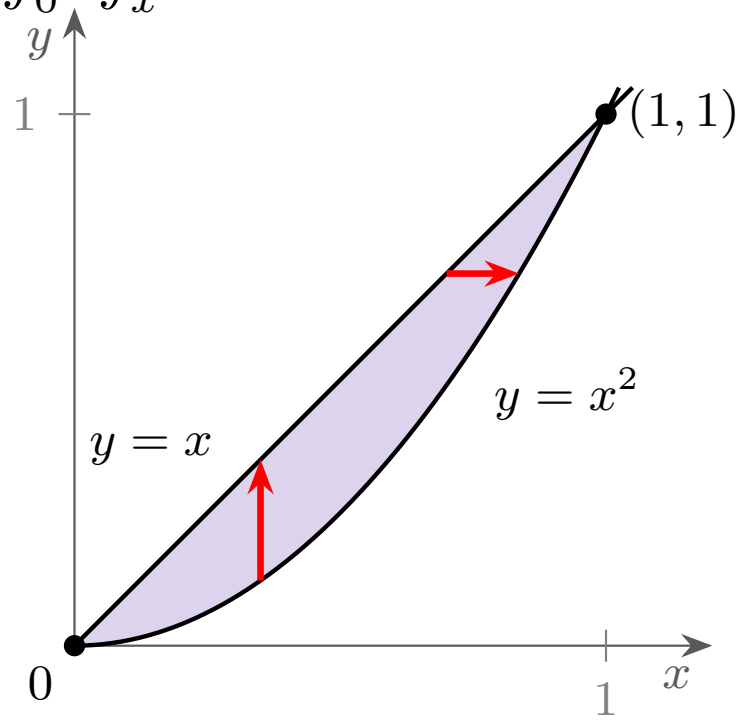
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$$\int_0^1 \int_{x^2}^{x} x \, dy \, dx = \int_0^1 \int_{y}^{\sqrt{y}} x \, dx \, dy$$

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Vertical slices ($dy\,dx$):

$$\int_0^1 \int_{x^2}^{x} x\,dy\,dx = \int_0^1 x(x-x^2)\,dx = \int_0^1 (x^2-x^3)\,dx = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

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Why Reverse? Example 4

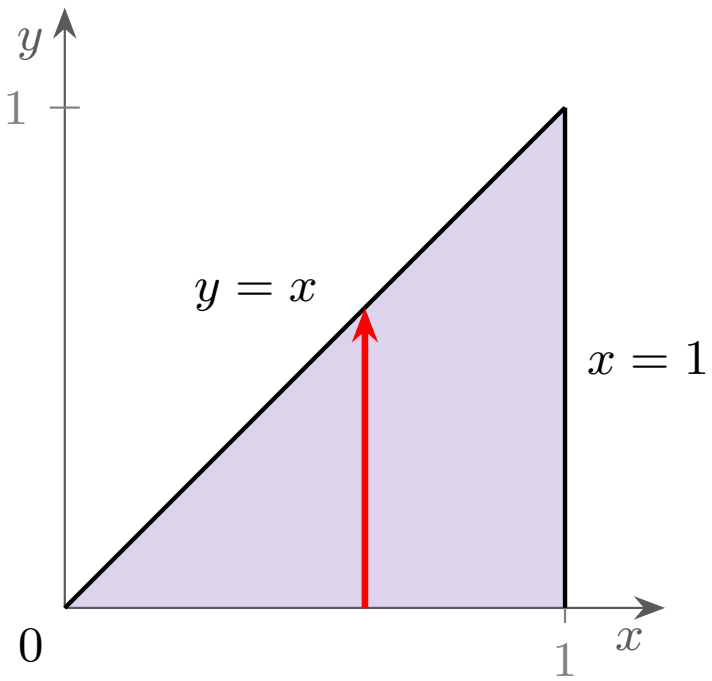
Evaluate $\displaystyle\int_0^1 \int_y^1 e^{x^2}\,dx\,dy.$

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Evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$. Problem: e^{x^2} has no elementary antiderivative!

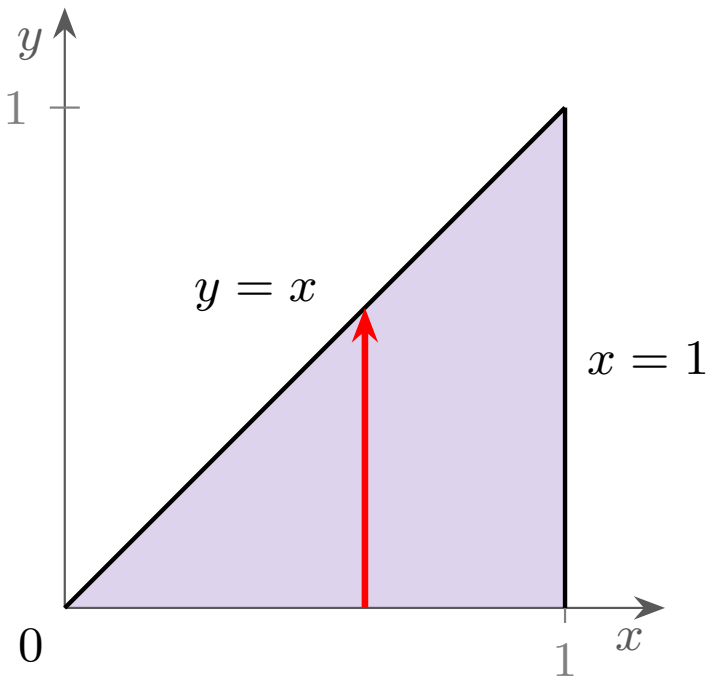
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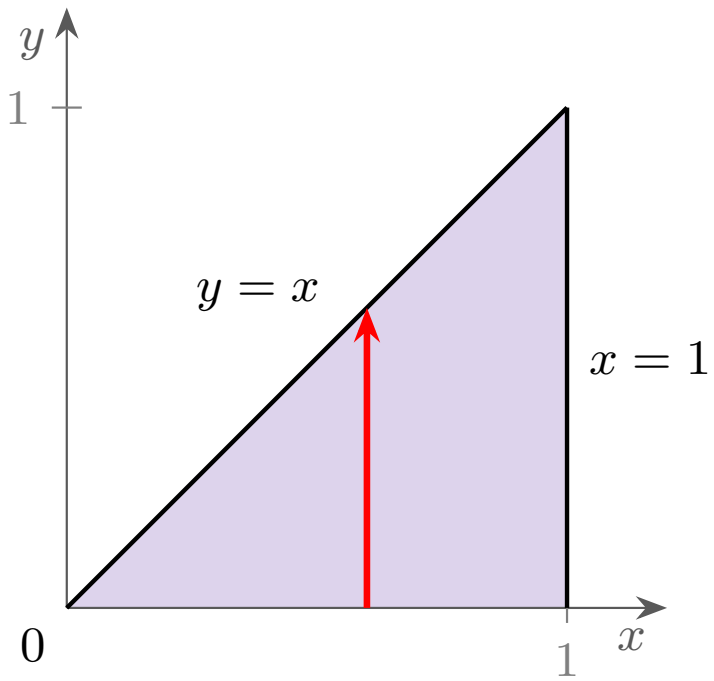
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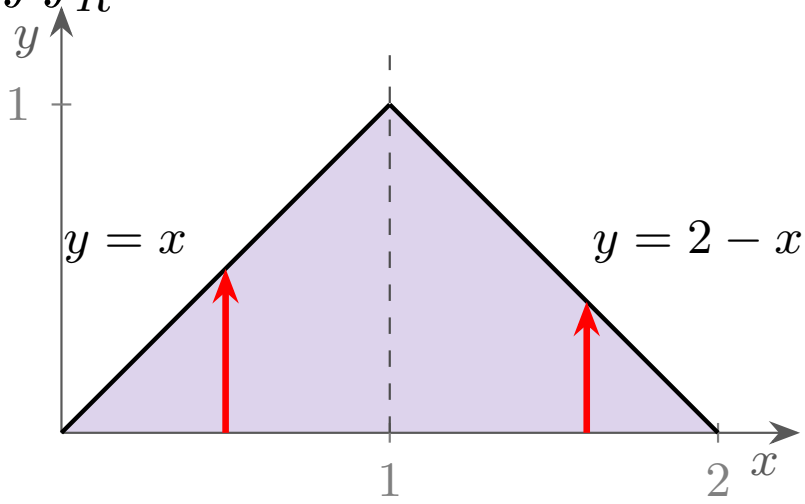


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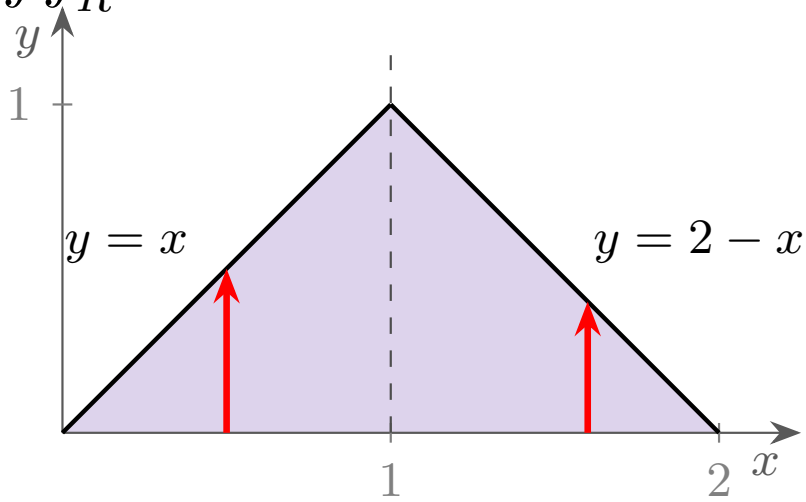
Example 5: When One Order Needs a Split

Let R be the triangle with vertices $(0,0)$, $(2,0)$, and $(1,1)$. Evaluate $\iint_R y\,dA$.



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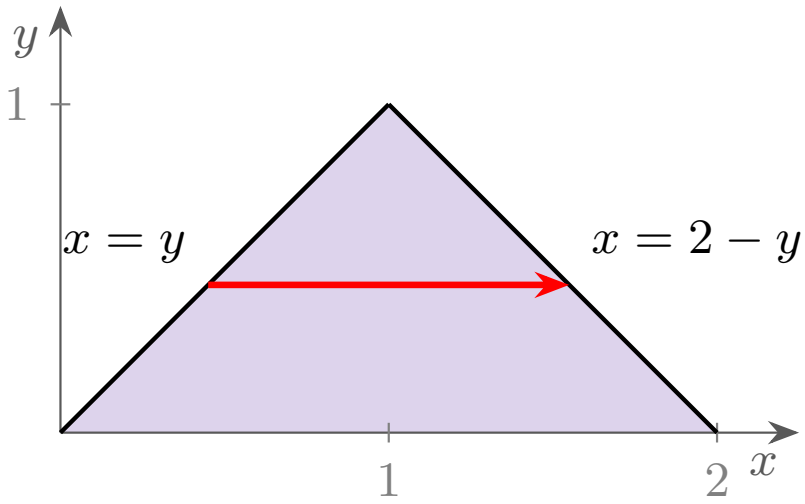
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Vertical slices: the top curve changes at $x = 1$, so we must split:

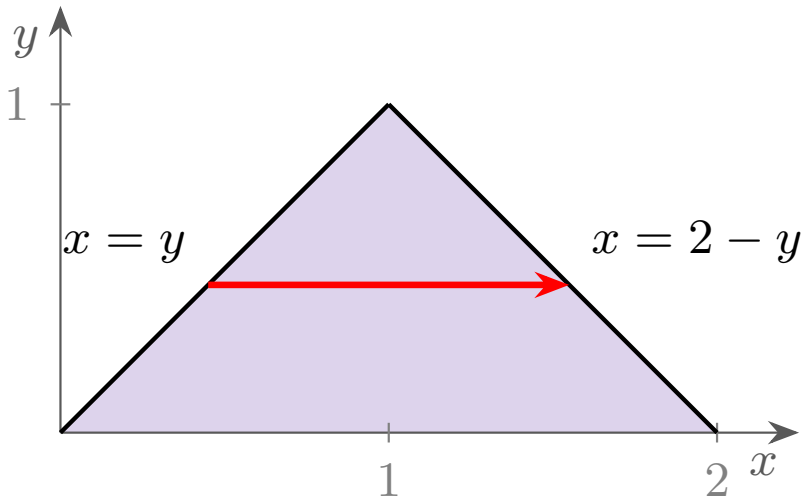
$$\int_0^1 \int_0^x y\,dy\,dx + \int_1^2 \int_0^{2-x} y\,dy\,dx$$
$$= \int_0^1 \frac{x^2}{2}\,dx + \int_1^2 \frac{(2-x)^2}{2}\,dx$$
$$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

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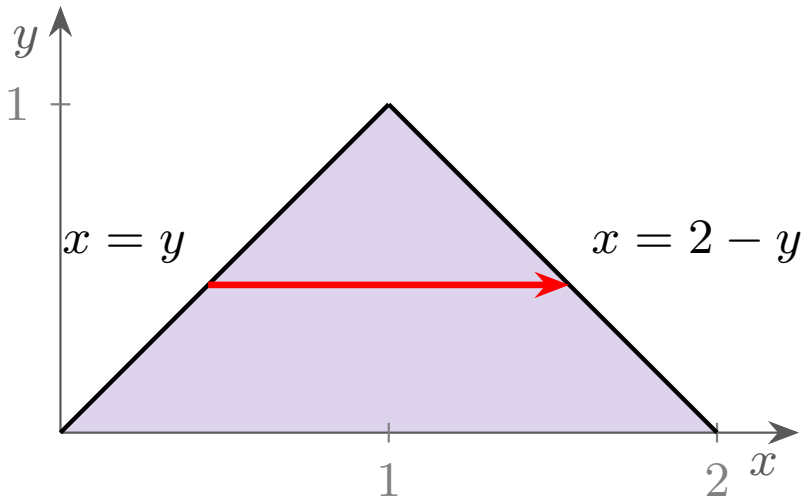


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Lesson: Before integrating, check whether one order gives a single integral. Choose that one.

Common Mistakes

- **Swapping bounds without re-deriving them.** $\int_0^1 \int_0^x f\, dy\, dx \neq \int_0^x \int_0^1 f\, dx\, dy$. The right side isn't even a number!

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- **Missing a split.** If a slice's entry or exit curve changes, you need more than one integral (or the other order).